

objective is to preserve the perovskite structure while swapping out different transition metal elements to create materials with a range of electronic and magnetic properties, including superconductivity, ferromagnetism and large thermopower. TMOs are useful in a range of applications from electronics to power transmission to refrigeration.

In an attempt to help scale down the size of electronic devices to atomic dimensions, researchers have demonstrated how to convert a particular transition metal oxide from a metal to an insulator by reducing its size to less than one nanometre thick. The study authors explain how they were able to synthesise atomically thin samples of lanthanum nickelate (LaNiO_3) utilising a precise growth technique known as molecular-beam epitaxy (MBE).

Researchers discovered that the process caused the material to abruptly change from metal to an insulator when its thickness was reduced to less than one nanometre. Following that change, the conductivity switched off, preventing electrons to flow through the material—a trait which could be beneficial for use in nanoscale switches or transistors.

Using a unique system that integrates the growth of MBE film with a method known as angle-resolved photoemission spectroscopy, researchers detailed how the specific movements and interactions of electrons in the material were altered, thus changing the thickness of their oxide films on an atom-by-atom basis. They found that once the films were less than three nickel atoms thick, electrons formed an unorthodox nanoscale pattern similar to that of a checkerboard. The discovery demonstrates the ability to control exotic transition metal oxides' electronic properties at the nanometre scale, while also revealing the surprisingly cooperative

interactions that rule electron behaviour in these types of extremely thin substances.

The authors report that their work helps pave the way for use of oxides in the creation of next-generation electronic devices. They wrote that these transition metal oxides have several advantages over conventional semiconductors, including the fact that their high carrier densities and short electronic length scales are desirable for miniaturisation, and that the strong interactions open new avenues for engineering emergent properties.

Bismuth compounds. These have become researchers' topological insulators of choice as they are simple

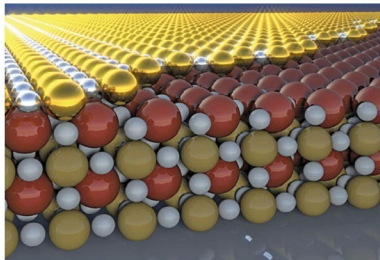
and economical to make. Chemists in China have precisely grown arrays of ultra-thin flakes of bismuth selenide and bismuth telluride on a surface.

Researcher's use a technique known as torque magnetometry to come up with first direct evidence to prove that samarium hexaboride is a topological insulator, observing oscillations in how the material responded to a magnetic field. Their technique also revealed that samarium hexaboride contains Dirac electrons on its surface. Because samarium hexaboride is a strongly correlated material, its electrons engage more closely with each other than in most solids, giving its interior electricity-blocking capabilities.

Spintronics uses electron spin at nanoscale to enable nanoscale magnetic sensors as well as advanced magnetic memories. Yet, researchers have not yet attempted to exploit new and exotic magnetic materials like (classical and quantum) spin ice—materials in which electron spin and orbital motion combine to produce very unusual effects such as magnetic monopoles and artificial photons.

Researchers have pioneered the physics and materials science of spin ice, recently creating the first atomic-scale thin films, as well as innovative new approaches to develop the spintronics of spin ice.

The team will investigate how electronic signals can be coupled to exotic magnetism—for example, currents of magnetic monopoles. The aim is to create new spintronic effects and new ways of probing exotic magnetism and how monopoles couple to electron-spins and photons using recently-developed spintronic probes. The successful application of spintronics to spin-ice thin films will start a new chapter in low-temperature condensed-matter physics and practical spintronics. **EFY**



Exotic material (Credit: redorbit.com)

and economical to make. Chemists in China have precisely grown arrays of ultra-thin flakes of bismuth selenide and bismuth telluride on a surface.

These bismuth compounds, researchers think, promise a new realm of fast, energy-efficient electronic devices and computers. Making well-defined nanoparticle arrays is a key step toward such devices. Growing uniform bismuth-based topological insulators at well-defined spots might be a step towards spin-based electronic devices that would perform computer logic using spin rather than electron charge to represent information, leading to faster, more energy-efficient computers.

Samarium hexaboride. An odd and iridescent material called samar-